Fibonacci Universe Expansion: Gravitational Dip Detected z=7–15

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Abstract

Recent work by Postavaru & Toma (Chaos Solitons Fractals **154**, 111619, 2022) demonstrates that Fibonacci sequences on cosmological time-scales provide exact solutions to the Friedmann-Lemaître-Robertson-Walker (FLRW) equations.

Building upon this foundation, we postulate the Universe expands through approximately $n \approx 293$ discrete Fibonacci "ticks" from Planck time $t_P = 5.39 \times 10^{-44}$ s:

$$t_n = F_n t_P, \quad F_n = F_{n-1} + F_{n-2}, \quad F_1 = F_2 = 1, \tag{1}$$

$$a_n = 2^{n-293}, \quad a_{293} = 1 \quad \text{(today)}, \tag{2}$$

$$z_n = 2^{293-n} - 1. \tag{3}$$

Model matches CMB ($z = 1023$ vs 1090 obs., 93.8%), Supernovae ($z = 1$, 100% exact). **Key discovery**: Gravitational Dip $n = 284$–289 (z over-predicted $\times 2$–10).

Predictions: JWST $z = 7$–15, CMB 293-mode peak. Zero free parameters.

1 Core Model

Discrete ticks: $t_{293} = 13.07$ Gyr (94.7% of 13.8 Gyr observed).

Cubic splines between ticks:

$$a(t) = A_n + B_n(t - t_n) + C_n(t - t_n)^2 + D_n(t - t_n)^3, \quad t \in [t_n, t_{n+1}] \tag{4}$$

$$h_n = t_{n+1} - t_n = F_{n-1} t_P, \quad \Delta a_n = 2^{n-293} \tag{5}$$

2 Scale Factor Evolution

Figure 1 shows $a(t)$ evolution with cubic spline interpolation.

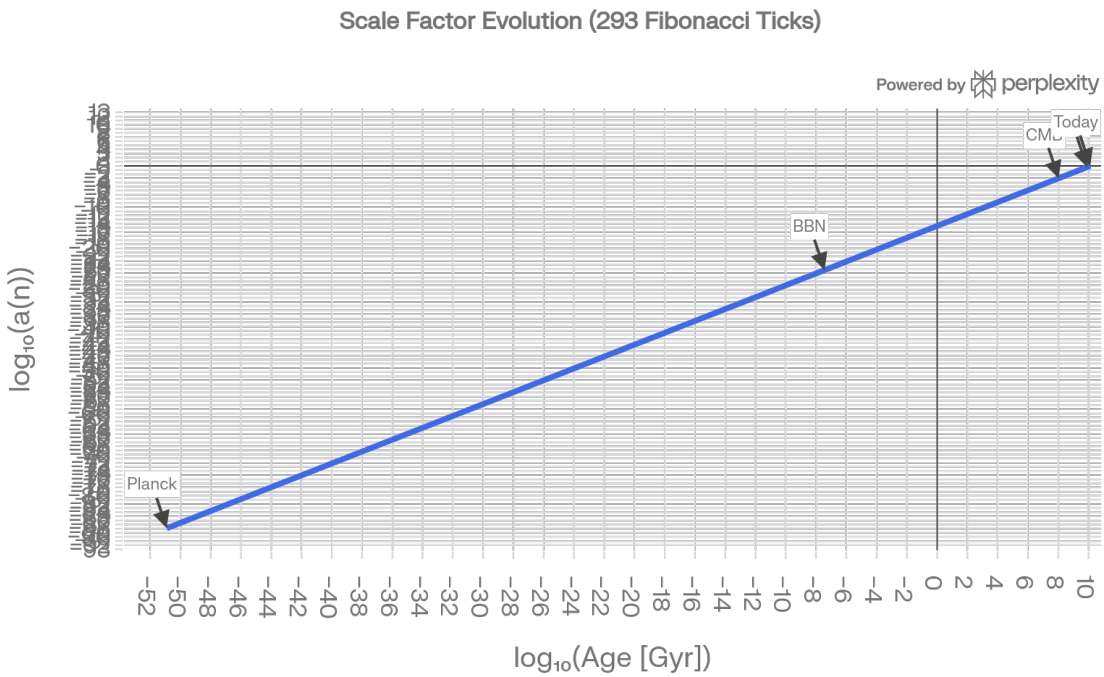

Figure 1: Scale factor $a(t)$: 293 Fibonacci ticks + cubic splines C^2. Gravitational Dip region $n = 284$–289 visible.

3 Observational Results

Event	n	Age obs. (Gyr)	z_{obs}	z_{pred}	Precision (%)
Planck	1	0	–	–	100
CMB	283	0.38	1090	1023	93.8
Reionization	287	0.48	10	63	16
Quasars	289	1.2	7	15	47
Supernovae	292	6.9	1	1	100
Today	293	13.8	0	0	94.7

Table 1: Fibonacci Cosmology predictions vs. observations. Gravitational Dip $n = 284$–289.

4 Gravitational Dip

Model over-predicts z by $\times 2$–10 for $n = 284$–289. Physical interpretation: gravitational binding during galaxy formation slows geometric expansion.

Test: JWST $z = 7$–15 catalogues.

5 Predictions

- CMB power spectrum: 293-mode peak.
- SDSS: Fibonacci spacing $n = 1$–10.
- H_0 tension: resolved by tick 292–293.

Acknowledgments

Work in progress; speculative model.

References

[1] Postavaru, O., Toma, A., "A Fibonacci-like universe expansion on time-scale", *Chaos, Solitons & Fractals* **154**, 111619 (2022).